

HW1- Computer Simulation II

Schewtschik, Guilherme (3748052)

April 19, 2026

1

In the first attempt, the Metropolis algorithm was used, but it became clear that for a lattice larger than 8×8 the run time around the critical temperature became too large, so instead the Wolff algorithm was chosen. With this we arrive at the following graph:

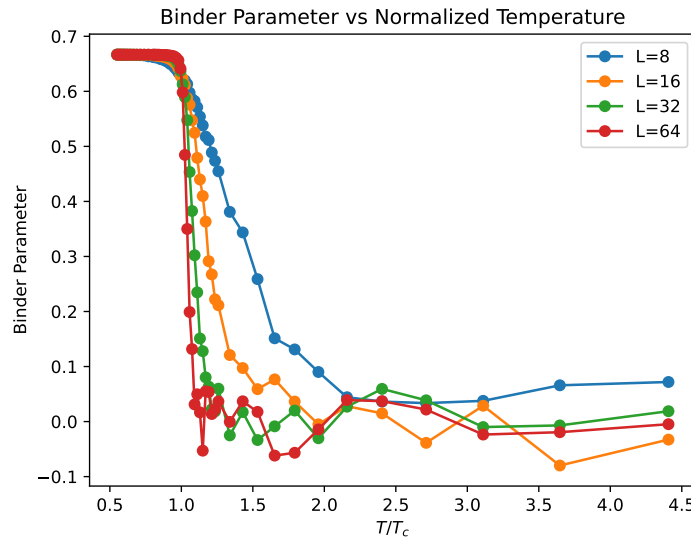


Figure 1: BinderXT

We can see that the value for U stays the similar for all L until reaching the critical temperature where they diverge.

Plotting the difference between the data points we get:

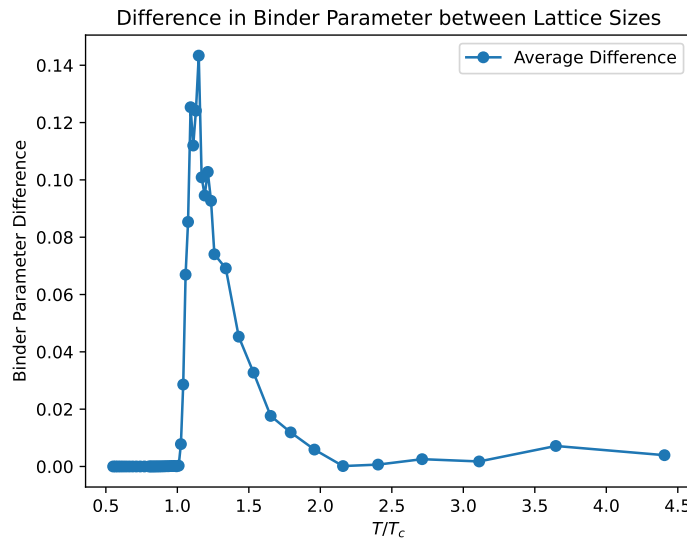


Figure 2: Caption

We can also see to a lower extent a convergence at higher temperatures.

The estimated value for U at T/T_c is:

Estimated U at T_c : $L=8$: 0.630, $L=16$: 0.628, $L=32$: 0.637, $L=64$: 0.641
Mean U at T_c : 0.634

2

Calculating the value of χ we can clearly see the peaks around the critical temperature:

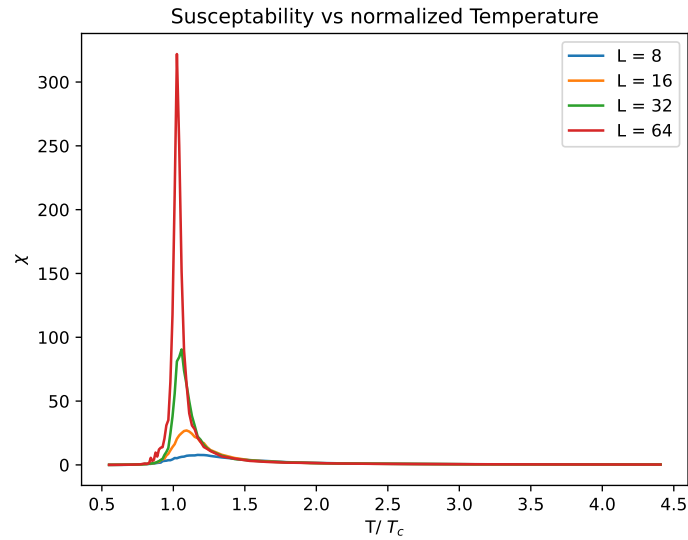


Figure 3: χ values

Taking the maximum of χ and plotting it against the lattice size L , we can see the relation between them appears to be exponential:

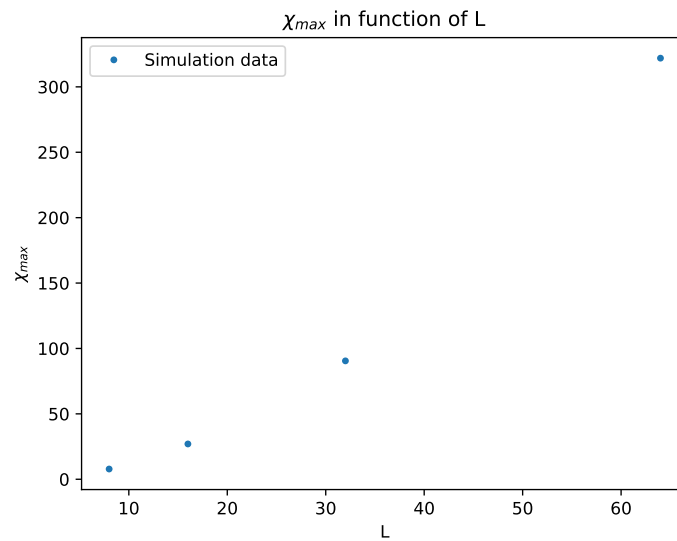


Figure 4: χ in function of L

For us to test the finite-size scaling Ansatz, we first need to make some adjustments, instead of using the given formula we shall work in log space:

$$\ln(\chi) = \frac{\gamma}{\nu} \ln(L) + c \quad (1)$$

Where c is a proportionality constant. Using this equation and fitting a line we can find the values for γ/ν and c .

$$\frac{\gamma}{\nu} = 1.783169... \quad c = -1.65374 \quad (2)$$

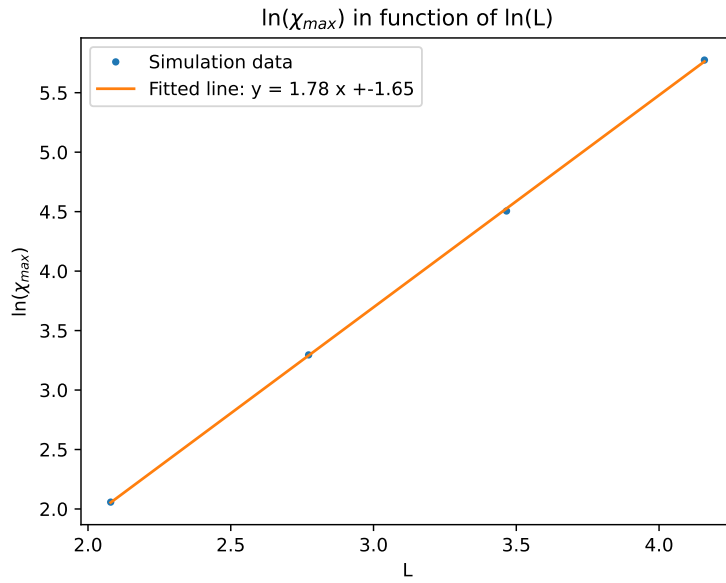


Figure 5: Fitted χ_{max}